St. John College of Engineering and Management, Palghar
Autonomous Institute
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DEPARTMENT OF COMPUTER ENGINEERING

MARUTI: AI Powered Voice Controlled Desktop Assistant

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Abstract

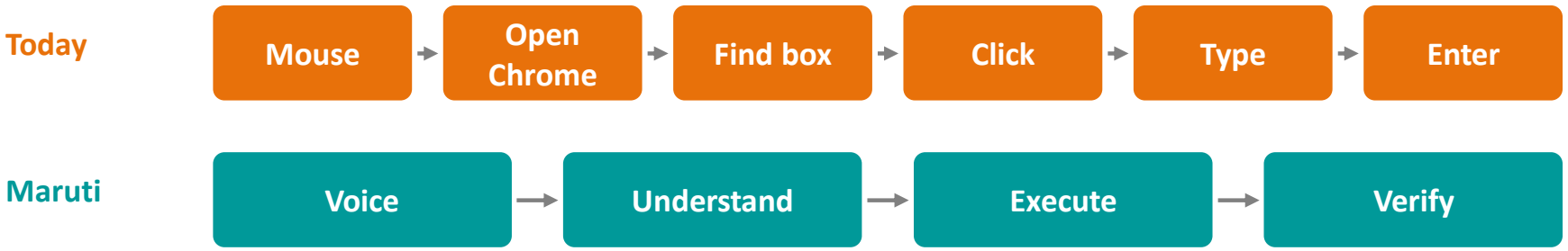
- **Problem:** Using a computer still depends on keyboard and mouse, which is hard for people with limited hand movement or injuries.
- **Approach:** Maruti turns speech into an intent, observes the screen using screenshots, OCR and AI vision, acts, and then verifies the result.
- **Outcome:** A working prototype for English, Hindi and Hinglish commands: open apps, search, type, scroll and click.
- **Motivation:** To make computers more natural, accessible and hands-free.

MVP pipeline



Problem Statement

- **Problem:** Simple tasks need many manual steps with keyboard and mouse.
- **Why it matters:** Users with limited hand movement, disabilities or injuries cannot work independently.
- **Relevance:** Supports accessibility, education and productivity.
- **Statement:** How can we make computer interaction natural, accessible and hands-free using voice and AI?



Objectives

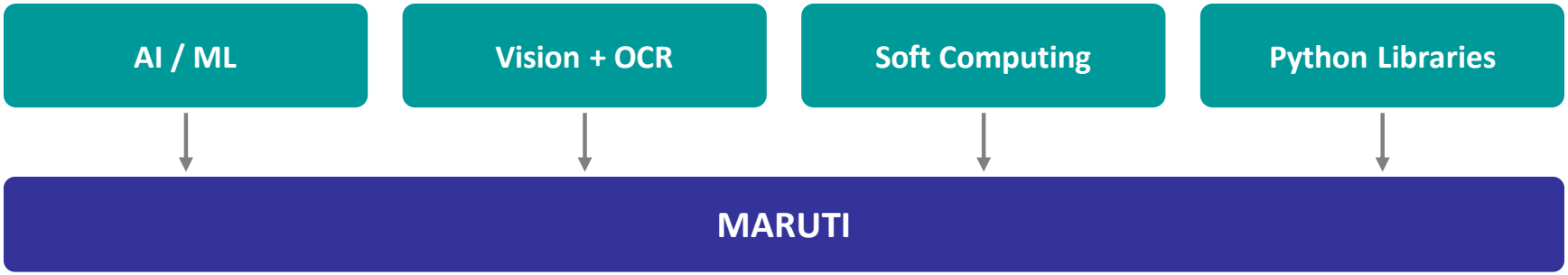
- Voice control in English, Hindi and Hinglish, measured with fixed test commands
- Separate conversation from computer tasks using intent routing
- Understand the screen with screenshot, OCR and AI vision to find click targets
- Perform actions (open, search, type, scroll, click) and verify each result
- Avoid unsafe actions: ask the user when confidence is low

Computer-use loop



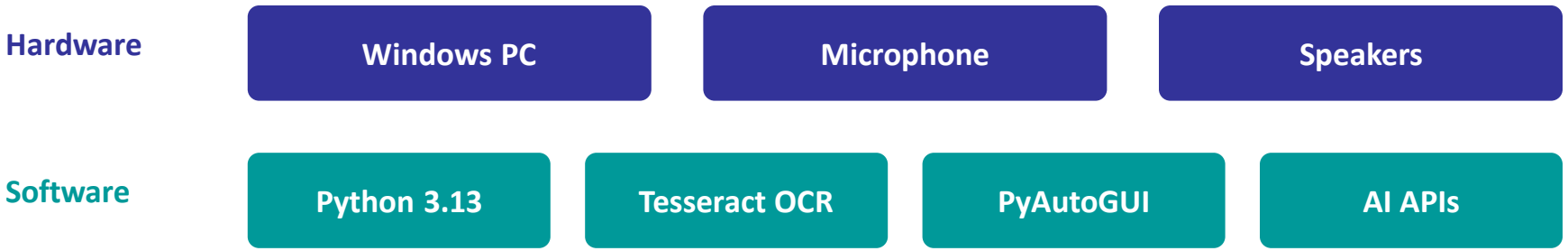
Background and Concepts

- **AI / ML basics:** speech recognition, intent understanding and multimodal (vision) models.
- **Computer vision and OCR:** reading on-screen text with coordinates to find click targets.
- **Soft Computing:** fuzzy logic for confidence-based decisions, fuzzy matching for uncertain text, neural networks.
- **Python essentials:** PyAutoGUI, SpeechRecognition, PyAudio, pytesseract, pywinauto, PySide6.

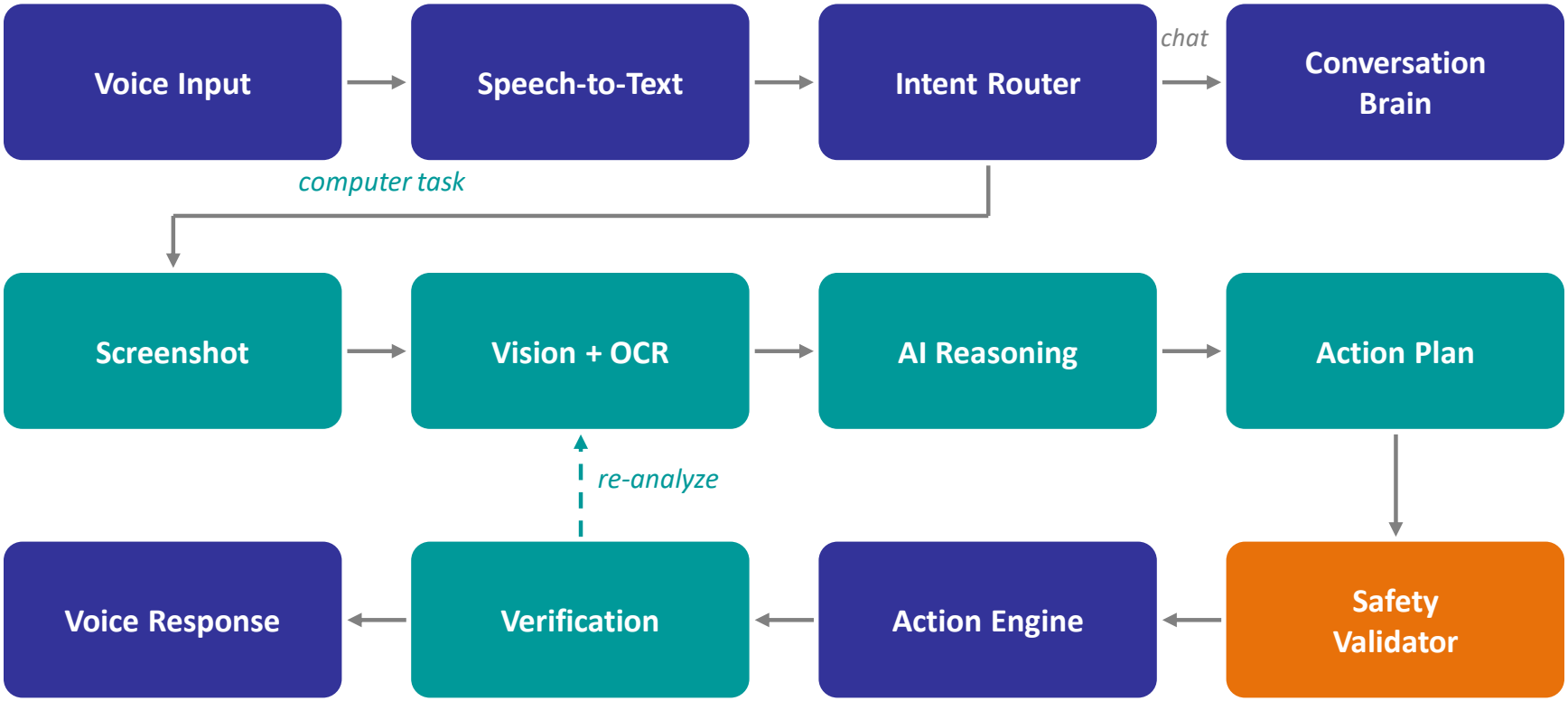


Requirements

- **Functional:** capture voice, recognise speech, detect intent, observe screen, act, verify, speak reply.
- **Hardware:** Windows 10/11 PC or laptop, microphone, speakers, internet.
- **Software:** Python 3.13, VS Code, PyAutoGUI, SpeechRecognition, PyAudio, Tesseract, pywinauto, AI APIs, PySide6.
- **Cost:** approx. ₹1,500 extra for the prototype; cloud AI/API usage is additional.



System Design and Architecture

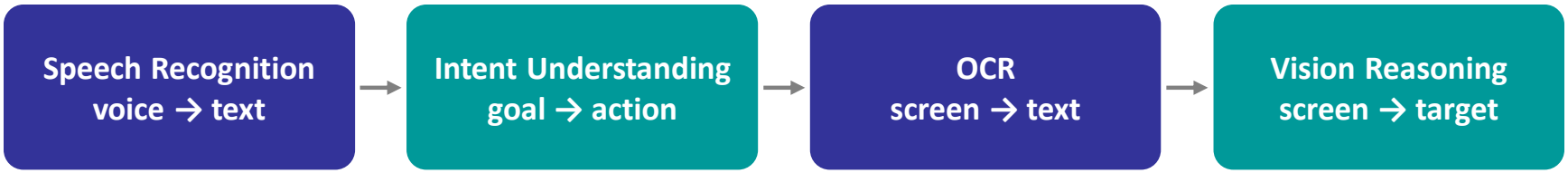


- Computer tasks run as a loop: act → new screenshot → verify → next step or re-analyze.
- If confidence is low Maruti asks; sensitive actions need confirmation.

AI/ML Techniques Used

- **Speech recognition:** converts English and Hindi speech to text so users can talk naturally.
- **Intent understanding:** decides chat or computer task, and the target app or action.
- **OCR:** reads on-screen text with coordinates to locate click targets.
- **Vision reasoning:** understands the screenshot to choose the next action (experimental).

How they work together



Soft Computing Design

Uncertain inputs from the AI / ML models

- **Speech text (may be wrong)**
- **OCR text (may be noisy)**
- **Vision result (has confidence)**
- **User command (informal / Hinglish)**

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Soft Computing layer (handles uncertainty)

Fuzzy logic	Fuzzy matching	Neural networks
High confidence: act Medium: ask the user Low: re-analyze	OCR “Logn” ≈ “Login” Alias “krom” ≈ Chrome	Inside the speech, OCR and vision models

↓

Safe decision: Act · Ask · Re-analyze

Design view: model outputs carry a confidence level that the soft-computing layer turns into a decision.

Implementation

- Voice recognition: English and Hindi
- Screenshot capture, OCR with coordinates
- Text-based clicking, Hindi Unicode typing, Enter, scroll
- Chrome control and intent routing (chat vs computer task)
- Validation and verification logic
- Multimodal visual reasoning: experimental

Core loop (pseudocode)

```
while not goal_done:
    shot   = take_screenshot()
    target = find_target(shot, goal)  # OCR + AI
    if confidence(target) < LIMIT:
        ask_user(); continue
    perform(target)   # click / type / scroll
    if verify(take_screenshot()):
        next_step()
```



Aligned SDG Point

- **SDG 10 – Reduced Inequalities:** lets people with limited hand movement or disabilities use computers independently.
- **SDG 9 – Industry, Innovation and Infrastructure:** applies AI, vision and automation to build an accessible software tool.
- **SDG 4 – Quality Education:** hands-free access to digital learning resources for more learners.

SDG 10
Reduced Inequalities

SDG 9
Industry, Innovation & Infrastructure

SDG 4
Quality Education

Conclusion

- **Summary:** Maruti turns a spoken goal into a verified computer action: understand, observe, act, verify.
- **Status:** voice, OCR, typing, scrolling and clicking are tested; multimodal reasoning and multi-step tasks are in development.
- **Impact:** more independent, hands-free computing and a base for a personalised assistant.
- **Future:** verified multi-step tasks, personalisation, Android and cross-platform support.



Q&A

Thank You!

Questions?

We would be glad to answer your questions. Thank you for your time and guidance.